# University of Colombo, Sri Lanka

*University of Colombo School of Computing*

UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING
29 SEP 2025
EXAMINATION & REGISTRATION

**BACHELOR OF SCIENCE IN COMPUTER SCIENCE**

**BACHELOR OF SCIENCE HONOURS IN COMPUTER SCIENCE**

**BACHELOR OF SCIENCE HONOURS IN SOFTWARE ENGINEERING**

**BACHELOR OF SCIENCE IN INFORMATION SYSTEMS**

**BACHELOR OF SCIENCE HONOURS IN INFORMATION SYSTEMS**

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Third Year Examination - Semester I - UCSC AY22 [held in September/October 2025]

SCS3203/IS3108 — Middleware Architecture

(Two (2) Hours)

**Answer ALL questions**

Number of Pages = 5 Number of Questions = 4

To be completed by the candidate

Index Number: | | | | | | | | |

055

**Important Instructions to candidates:**

I. Students should answer in the medium of English language **only using the answer books/sheets provided**.

II. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.

III. This paper consists of 4 questions in 5 pages (including the Cover Page).

IV. **Answer all questions.** All questions carry equal marks **(25 marks).**

V. Programmable Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are **not allowed**.

VI. Non-Programmable calculators are **not allowed.**

VII. Do not tear off any part of this answer book. Under no circumstances may this book, used or unused, be removed from the Examination Hall by a candidate.

## Question 1

(a) Explain the core difference between the Proxy and Service Locator architectural patterns and illustrate with a scenario where they are combined to provide location transparency for a client application in a distributed system.

*[07 Marks]*

(b) Explain the roles of the three (03) main components of the Model-View-Controller (MVC) architectural pattern and describe how MVC can be integrated into a 3-Tier architecture to build a multi-tiered, thin-client application.

*[06 Marks]*

(c) Reliability and availability are two crucial metrics for evaluating fault tolerance. Given two platforms: Platform A with an MTTF (Mean Time To Fail) of 1 year and an MTTR (Mean Time To Repair) of 48 hours, and Platform B with an MTTF of 6 months and an MTTR of 2 hours, discuss the relative differences in their availability and reliability.

*[06 Marks]*

(d) Discuss three (03) key differences between Service-Oriented Architecture (SOA) and Microservices Architecture (MSA) by comparing them across the following dimensions:

(i) Service Granularity
(ii) Data Persistence
(iii)Deployment

*[06 Marks]*

## Question 2

A principal architect is tasked with modernizing a digital banking platform. The existing system is a single, large monolithic application that handles all services, including customer accounts, credit management, loan processing, and transaction history. The platform is experiencing several issues, including slow development cycles, difficulty scaling individual services, and a single point of failure.

(a) To address the issues, the architect adopts a microservices approach. They choose an incremental strategy to avoid a risky "Big Bang" rewrite. Explain how the "Stop Digging" strategy can be used to introduce new functionality, like a mobile wallet service, into the existing platform.

*[04 Marks]*

(b) The architect decides to use Polyglot Persistence for the microservices. The new mobile wallet service will use a NoSQL database for its transactions, while the legacy loan processing service will continue to use a relational database.

(i) Explain the benefits of using the Polyglot Persistence approach in this scenario.

*[03 Marks]*

(ii) Discuss how an Event-Driven Architecture (EDA) can be used to handle a complex business transaction, such as reserving credit for a new mobile wallet payment, which involves both the legacy credit service and the new mobile wallet service.

*[05 Marks]*

(c) The digital banking platform uses microservices architecture. To handle communication from its mobile client, the principal architect must decide on the best method for the client to access the various microservices. List down different methods a mobile client can use to access microservices and justify which method is most suitable for this scenario, especially as the platform undergoes an incremental migration from a monolith using the "Stop Digging" strategy. Your justification should explain how the chosen method solves the problem of a mismatch between the needs of the client and the fine-grained APIs of the microservices.

*[07 Marks]*

(d) The microservices architecture uses several client-service interaction styles to enable the most suitable engagement based on the nature of the interaction. The two key dimensions for client-service interaction styles are: number of participants (one-to-one and one-to-many) and communication mode (synchronous and asynchronous).
Select the most suitable client-service interaction style for each of the four (04) client and service interaction scenarios given in the table below and justify your selection for each scenario.

| Client | Service |
|---|---|
| Client request to add a new user to the system. | Service to validate the user and save the user profile. The client has to wait until the new user profile is successfully saved. |
| Client request to update customer contact details. | Customer contact service to validate and update the contact details. The client is not waiting for a response and can proceed with its other business logic. |
| A new product is added to the catalogue by an administrator. | All active users receive a push notification about the new product. |
| Client request to retrieve a customer's order history. | Order service and other related services to aggregate a complete order history. The client is not bound by time and expects a response, but it can be delayed. |

*[06 Marks]*

**Question 3**

(a) "SmartLogistics" is a large-scale logistics company with operations spanning multiple countries. The company has a diverse set of IT systems, including a legacy system for warehouse management, a cloud-based application for fleet tracking, and a third-party application for customs documentation. The company wants to integrate these disparate systems to provide a unified platform for tracking a shipment from origin to destination. The architect is evaluating two potential architectural patterns for this modernization: Service-Oriented Architecture (SOA) and Microservices Architecture (MSA).

(i) Given the company's need to integrate diverse, heterogeneous systems and the presence of legacy components, select the most suitable architectural pattern between SOA and MSA. Justify your selection by discussing the strengths of the chosen architecture in this specific context.

*[06 Marks]*

(ii) A single customer requests to track a shipment requires calling multiple services to get information from the warehouse, fleet tracking, and customs systems. Propose a combination of integration patterns that ***Enterprise Service Bus (ESB)*** would use to fulfil this request. Depict the connectivity and dependencies of these patterns in a simple diagram.

*[08 Marks]*

(b) A global financial services company, "FinCorp," has a complex IT landscape. Its core banking system, built in the early 2000s, uses Remote Method Invocation (RMI) for inter-module communication within its Java-based application. Some of its subsidiary systems, acquired through mergers, use Common Object Request Broker Architecture (CORBA) to expose their services. FinCorp is now undertaking a major initiative to adopt modern Service-Oriented Architecture (SOA) to consolidate its services and improve agility.

(i) Explain the key differences between the RMI and CORBA architectures, focusing on language dependency and interoperability.

*[05 Marks]*

(ii) Given FinCorp's existing landscape of RMI and CORBA components, discuss how SOA middleware, specifically the Enterprise Service Bus (ESB), would be a more suitable choice for its modernization goals compared to these legacy technologies.

*[06 Marks]*

**Question 4**

"TelcoCo," a major telecommunications provider, is modernizing its core service platform by adopting microservice architecture. The company has decided to use Docker for containerization and Kubernetes for container orchestration to improve agility, scalability, and resource utilization.

(a) With TelcoCo's services deployed as independent containers, managing transactions across different services is a key challenge. In a monolithic application, two-phase locking (2PL) is a common method for handling concurrent transactions. However, this is not suitable for a distributed microservices environment that uses asynchronous, event-driven communication. Discuss the key difficulties of using traditional 2PL in this environment and propose an alternative transaction management pattern suitable for event-driven microservices.

*[06 Marks]*

(b) Explain three (3) advantages of using containerization over traditional virtualization (e.g., Virtual Machines) for deploying the microservices on TelcoCo's platform.

*[06 Marks]*

(c) Describe the primary function of Kubernetes in this containerized environment and justify why it is a critical technology for the implementation of this scale with hundreds of microservices.

*[06 Marks]*

(d) Fault tolerance is a crucial aspect of a highly available system. TelcoCo must ensure its services are resilient to failures. Explain the method of fault-masking using Triple Modular Redundancy (TMR) to achieve high availability and describe the application of TMR to a mission-critical system, such as a real-time call rating service in a telco billing platform.

*[07 Marks]*

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